Extragalactic Astrophysics and Galaxy Formation

Master on Astrophysics, Particle Physics and Cosmology





Course Structure

- Goal: To give the present view of the structure and dynamics of galaxies as well as their formation and evolution in a cosmological context, paying especial attention to the physical mechanisms involved

- Course format: Four 60-min lectures per week (Spring semester)

- Blocks:
  - Extragalactic astrophysics: Dr. J.M. Solanes (18 sessions)
  - Galaxy formation and evolution: Dr. A. Manrique (18 sessions)
  - Simulations of large-scale structure & galaxy formation: Dr. C. Laporte (7 sessions)

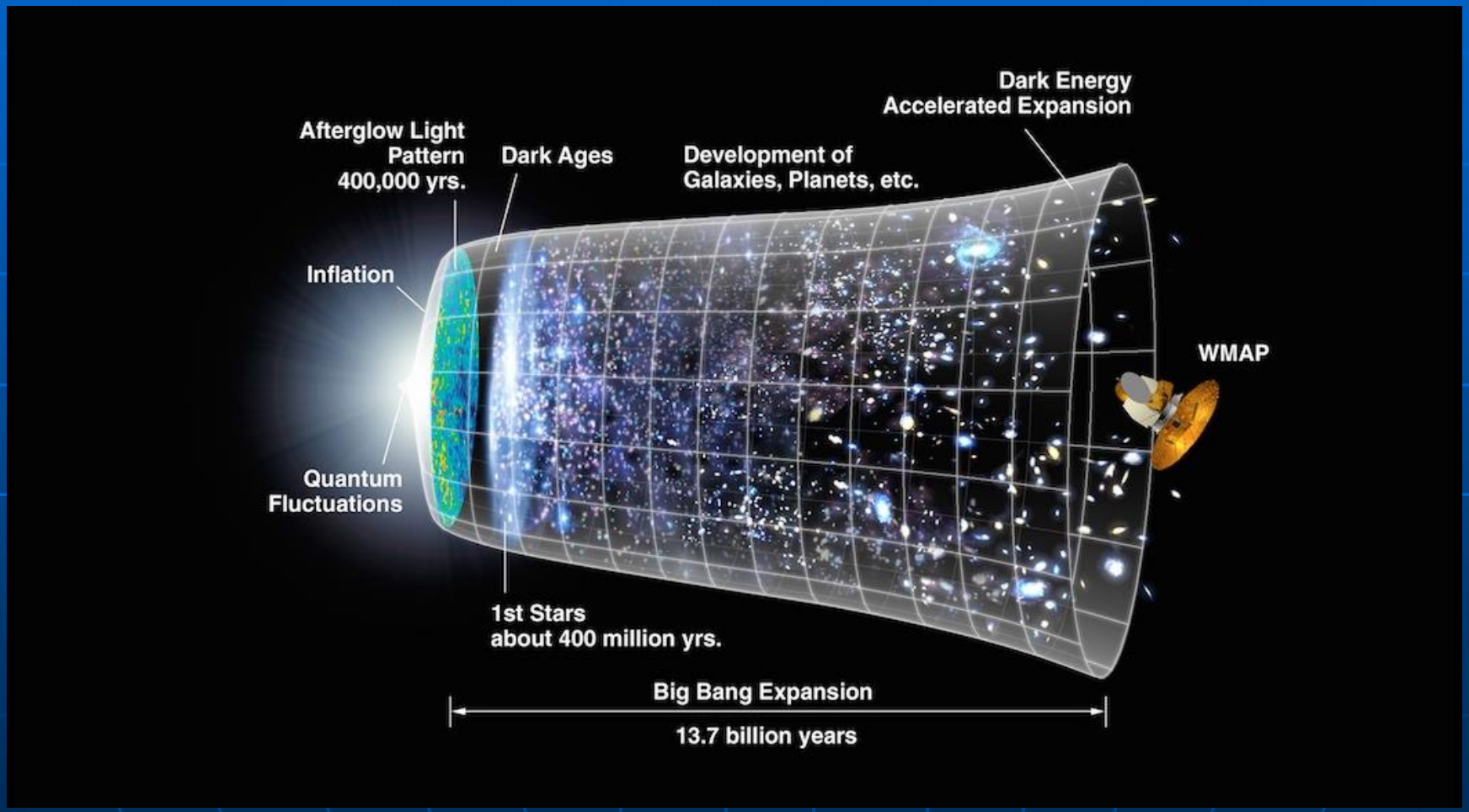
Extragalactic Astrophysics



Contents: Part I

1. PRELIMINARIES.
2. INTRODUCTION TO GALAXIES.
3. ACTIVE GALACTIC NUCLEI (AGN).
4. SPIRAL GALAXIES (LTGs).
5. ELLIPTICAL GALAXIES (ETGs).
6. GALAXY GROUPS AND EVOLUTION.

Galaxy Formation

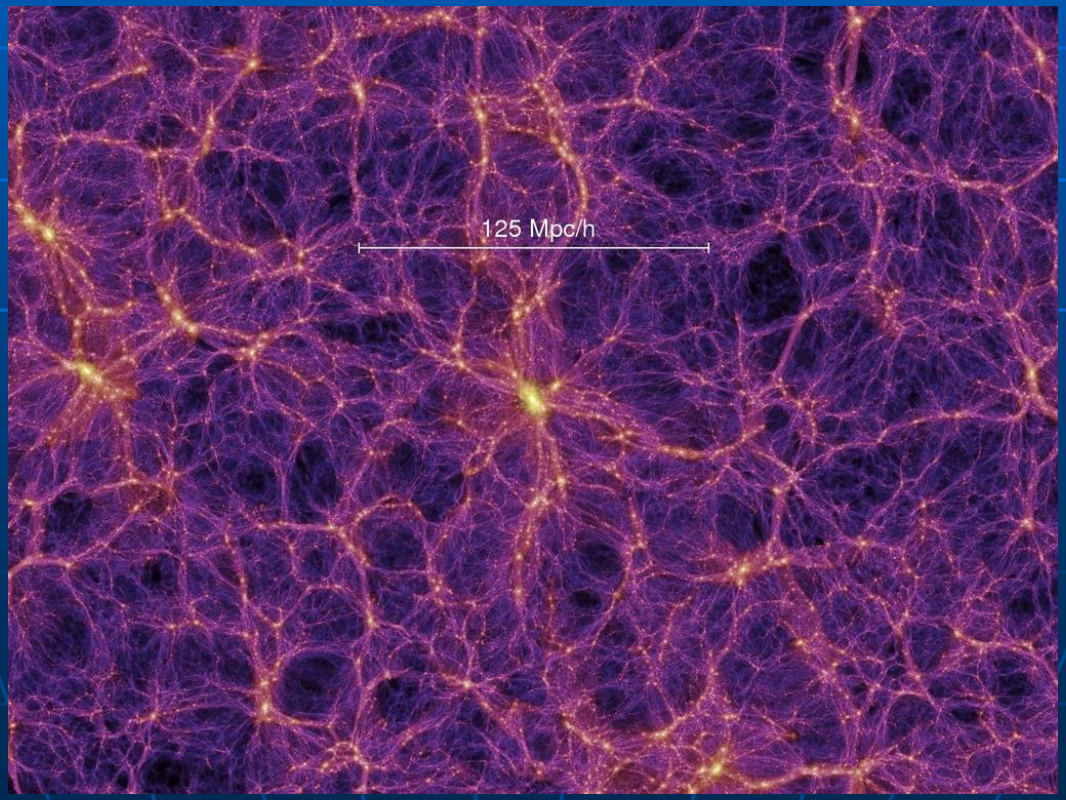


Contents: Part II

1. STRUCTURE FORMATION IN THE UNIVERSE.
2. COSMIC DENSITY PERTURBATIONS: LINEAR EVOLUTION.
3. SPHERICAL COLLAPSE.
4. RELAXATION MECHANISMS AND TIME SCALES.
5. DARK MATTER HALOS.
6. GALAXY FORMATION AND EVOLUTION.
7. THE HIGH-REDSHIFT UNIVERSE.

Contents: Part III

1. MODELING LARGE-SCALE STRUCTURE AND GALAXY FORMATION: SIMULATIONS.



Course Grading

- It will be based 75% on a research work: 2-student groups will prepare (50%) and present 25%) a meeting-like poster with the results of the analysis of a galaxy cluster.
- The poster (in English) will be presented and the results explained (in English) in an oral session at the end of the semester (early July).
- The contents of the poster and the oral presentation will be evaluated according to grading sheets (similar to TFM).

Course Grading

- The other 25% of the grade will be based on the evaluation of a series of tasks that will be proposed through the Virtual Campus.

Course Material

- UB *Campus* Virtual:
  - lecture notes
  - tasks, galaxy cluster project
  - recommended readings (e.g. review papers on specific topics)
- Bibliography (textbooks):
  - GALACTIC DYNAMICS (2on. Ed.), Binney & Tremaine, Princeton University Press (2008)
  - GALAXY FORMATION AND EVOLUTION, H. Mo, S.D.M. White & F. van den Bosch, Cambridge University Press (2008)
  - GALAXY FORMATION (2on. Ed.), M.S. Longair, Springer (2008)
  - GALAXIES IN THE UNIVERSE (2on. Ed.), Sparke & Gallagher, Cambridge University Press (2007)
  - GALAXY FORMATION AND EVOLUTION, H. Spinrad, Springer (2005)